STUDENT MANAGEMENT SYSTEM DATABASE DOCUMENTATION

1. ***Table Structure***

Departments Table

COLUMN NAME	DATA TYPE	FEATURE	DESCRIPTION
DepartmentID	VARCHAR(30)	Primary Key	Unique ID for each department
DepartmentName	VARCHAR(50)	NOT NULL	Name of the Department

Students Table

COLUMN NAME	DATA TYPE	FEATURE	DESCRIPTION
StudentID	VARCHAR(30)	Primary Key	Unique ID for each student
FirstName	NVARCHAR(30)	NOT NULL	Student’s first name
LastName	NVARCHAR(30)	NOT NULL	Student’s last name
Gender	VARCHAR(10)	NOT NULL	Gender of the student
Date_of_Birth	DATE	NOT NULL	Student’s birth date

Courses Table

COLUMN TABLE	DATA TYPE	FEATURE	DESCRIPTION
CourseID	VARCHAR(30)	Priimary Key	Unique ID for each course
CourseName	NVARCHAR(100)	NOT NULL	Name of the course

Instructors Table

COLUMN TABLE	DATA TYPE	FEATURE	DESCRIPTION
InstructorID	VARCHAR(30)	Primary Key	Unique ID for each instructor
Title	VARCHAR(10)	NOT NULL	Instructor’s title
FirstName	NVARCHAR(30)	NOT NULL	Instructor’s first name
LastName	NVARCHAR(30)	NOT NULL	Instructor’s last name
Gender	VARCHAR(10)	NOT NULL	Instructor’s gender

Enrollments Table

COLUMN TABLE	DATA TYPE	FEATURE	DESCRIPTION
EnrollmentID	VARCHAR(30)	Primary Key	Unique ID for each enrollment
StudentID	VARCHAR(30)	Foreign Key	References Students(StudentID)
CourseID	VARCHAR(30)	Foreign Key	Reference Courses(CourseID)
InstructorID	VARCHAR (30)	Foreign Key	Reference Instructors(InstructorID)
DepartmentID	VARCHAR(30)	Foreign Key	Reference Departments(DepartmentID)
EnrollmentDate	DATE	NOT NULL	Date of Enrollment

2. Table Relationship

Students ↔ Enrollments

One to many Relationships

- One Students can have many enrollments
- Each enrollments belong to one students

Courses ↔ Enrollments

One to many Relationships

- Each course can have many students enrolled
- One course can appear in many enrollments

Departments ↔ Enrollments

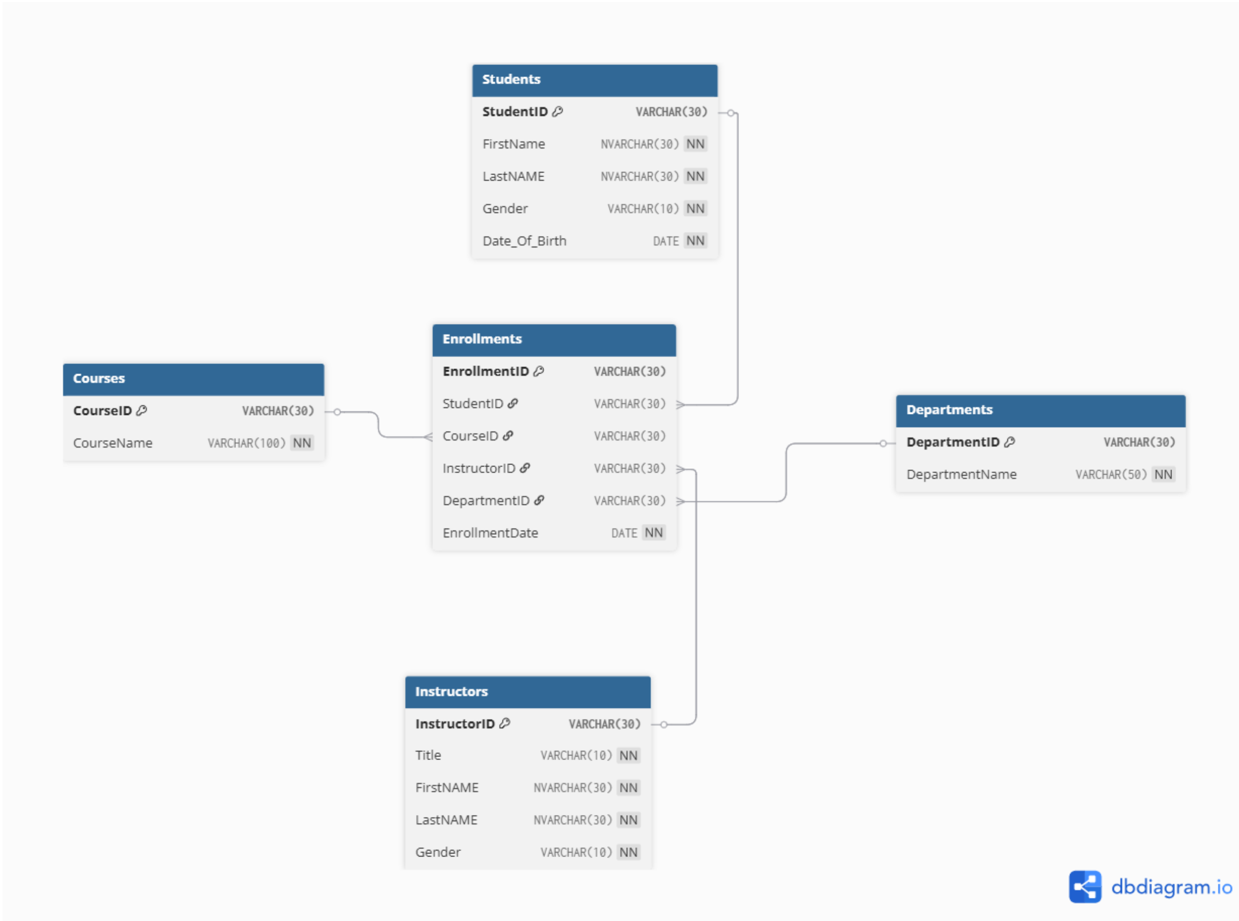
One to many Relationships

- Each departments can have many students enrolled
- One department can have many enrollment

Instructors ↔ Enrollments

One to many Relationships

- One instructor can teach many enrollments



3. QUERY

--How many students are enrolled in each course

```
SELECT c.CourseName, COUNT(e.StudentID) AS TotalStudentsEnrolled
FROM Courses c
LEFT JOIN Enrollments e ON c.CourseID = e.CourseID
GROUP BY c.CourseName
ORDER BY TotalStudentsEnrolled DESC;
```

```sql
--Which students are enrolled in more than two courses
SELECT  s.StudentID,  s.FirstName, s.LastNAME,  COUNT(e.CourseID) AS TotalCourses
FROM Students s
JOIN Enrollments e ON s.StudentID = e.StudentID
GROUP BY s.StudentID, s.FirstName, s.LastNAME
HAVING COUNT(e.CourseID) > 2
ORDER BY TotalCourses DESC;

--list all students who are not enrolled in any course
SELECT   s.StudentID, s.FirstName,  s.LastNAME,  s.Gender, s.Date_Of_Birth
FROM Students s
LEFT JOIN Enrollments e ON s.StudentID = e.StudentID
WHERE e.StudentID IS NULL;

--which course has the highest number of of enrollments
SELECT TOP 1  c.CourseName,  COUNT(e.StudentID) AS TotalEnrollments
FROM Courses c
JOIN Enrollments e  ON c.CourseID = e.CourseID
GROUP BY c.CourseName
ORDER BY TotalEnrollments DESC;

--what is the total number of credits students are taking
SELECT  SUM(c.Credits) AS TotalCreditsTaken
FROM Enrollments e
JOIN Courses c ON e.CourseID = c.CourseID;
```

```sql
--what is the gender distribution of students

SELECT  Gender, COUNT(*) AS TotalStudents

FROM Students

GROUP BY Gender;


--who are the students enrolled in a specific course(=Mathematics)

SELECT   s.StudentID,  s.FirstName,  s.LastNAME,  c.CourseName

FROM Students s

JOIN Enrollments e  ON s.StudentID = e.StudentID

JOIN Courses c ON e.CourseID = c.CourseID

WHERE c.CourseName = 'Mathematics'

ORDER BY s.LastNAME, s.FirstName;


--which instructor teaches the most enrolled students

SELECT   i.InstructorID, i.Title, i.FirstNAME,  i.LastNAME,  COUNT(e.StudentID) AS
TotalStudentsTaught

FROM Instructors i

JOIN Enrollments e ON i.InstructorID = e.InstructorID

GROUP BY i.InstructorID, i.Title, i.FirstNAME, i.LastNAME

ORDER BY TotalStudentsTaught DESC;


--how many students are enrolled in each department

SELECT d.DepartmentName,  COUNT(e.StudentID) AS TotalStudents

FROM Departments d

LEFT JOIN Enrollments e  ON d.DepartmentID = e.DepartmentID

GROUP BY d.DepartmentName
```

```sql
ORDER BY TotalStudents DESC;


--checking the whole database structure

SELECT  e.EnrollmentID,  s.StudentID, s.FirstName, s.LastNAME, s.Gender,  c.CourseName,

   i.Title + ' ' + i.FirstNAME + ' ' + i.LastNAME AS InstructorName, d.DepartmentName,
e.EnrollmentDate

FROM Enrollments e

JOIN Students s ON e.StudentID = s.StudentID

JOIN Courses c ON e.CourseID = c.CourseID

JOIN Instructors i  ON e.InstructorID = i.InstructorID

JOIN Departments d ON e.DepartmentID = d.DepartmentID

ORDER BY e.EnrollmentDate;
```